

Week 4 cheat sheet: tests, effect size and power

Quantitative Methods in Neuroscience, 2026-27. Keep it next to you while you work.

The words

H_0	the null: there is no effect
H_1	the alternative: there is one
α	the risk of a false alarm you accept, usually 0.05
p-value	how often chance alone would give a result this extreme, <i>if</i> H_0 were true
type I	rejecting a true null, a false alarm
type II	missing a real effect
power	1 minus type II, the chance of finding a real effect

What a p-value is not

It is not the probability that the null is true, and not the size of the effect. A small p from a small sample says "worth a second look", not "settled".

The three tests

```
from scipy import stats

# two independent groups, the default choice
stats.ttest_ind(a, b, equal_var=False) # Welch

# the same subjects measured twice
stats.ttest_rel(before, after)

# one group against a fixed value
stats.ttest_1samp(x, popmean=0.5)
```

Each returns the statistic and the p-value, two-sided.

The statistic

A difference, divided by how much a difference of that size wanders by chance:

```
t = (mean_a - mean_b) / standard_error
```

Bigger sample, smaller standard error, larger t for the same difference.

Assumptions

- **Independence.** One number per mouse, never one per trial.
- **Normality** of the quantity being averaged, or a sample large enough for the CLT.
- **Equal variance** for the pooled version. Welch does not need it, which is why it is the default.

```
stats.levene(a, b) # do the two spreads differ?
```

Effect size

```
sd_pooled = np.sqrt(((na - 1) * va + (nb - 1) * vb) / (na + nb - 2))
d = (mean_a - mean_b) / sd_pooled
```

Cohen's d is the difference in standard deviations, so it does not grow with the sample the way t does. Roughly: 0.2 small, 0.5 medium, 0.8 large.

Confidence interval

```
t_crit = stats.t.ppf(0.975, df)
low = diff - t_crit * se
high = diff + t_crit * se
```

The range of differences the data do not rule out. It says more than a p-value, because its width shows how much doubt is left.

Distributions by hand

stats.t.sf(x, df)	area in the upper tail, double it for two-sided
stats.t.ppf(q, df)	the inverse, where critical values come from
stats.t.pdf(x, df)	the curve itself, for plotting
stats.norm	the same three, for the Normal

Power, by simulation

```
hits = 0
for _ in range(2000):
    a = rng.normal(mu_a, sd, size=n)
    b = rng.normal(mu_b, sd, size=n)
    if stats.ttest_ind(a, b, equal_var=False).pvalue < 0.05:
        hits = hits + 1
power = hits / 2000
```

No formula to remember: invent the experiment many times and count how often it works.

Several tests at once

Run m tests at $\alpha = 0.05$ and the chance of at least one false alarm is far above 5%. The blunt fix is **Bonferroni**: compare each p against α / m , and say in the text how many tests you ran.

Reporting a comparison

- the design, and what one observation is
- the sample size in each group
- the null, and one-sided or two-sided
- which test, and why that one
- the statistic, the degrees of freedom, the p-value
- the effect size and a confidence interval
- one sentence of plain language saying what it means